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Summary

	Total Possible Marks	Your Marks
Lesson 1	23	
Lesson 2	29	
Lesson 3	22	
Lesson 4	26	
TOTAL	100	

Teacher's Comments

Teacher's Signature

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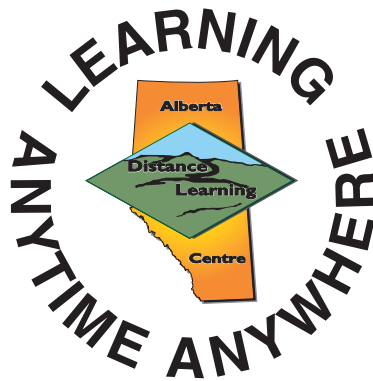
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Forensic Science 35

Assignment Booklet 1

Forensic Toxicology



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Forensic Science 35

Assignment Booklet 1

Assessment

For successful completion of this course, you must

- 1. Complete all questions in each Assignment Booklet to the best of your ability.**
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- 2. Achieve at least 40% on the final exam.**
In accordance with Alberta Distance Learning Centre policy, the final exam is worth 50% of the final course mark.
- 3. Achieve a final course mark of at least 50%.**

Process

- **Read the course material** before working on the activities in this Assignment Booklet. The *Forensic Science 35 Student Module Booklets* contain the information necessary for successful completion of your work.
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Module 1, Lessons 1 – 4

Total
23

Lesson 1 Assignment: The Effects of Illegal Drugs upon the Human Body

Value
10

Part A: Multiple-Choice Questions

Decide which choice best completes the statement or answers the question. Print the letter that corresponds to your choice on the blank next to the question.

- A** 1. Forensic toxicology helps solve crimes by detecting and identifying
- A. drugs and poisons
 - B. pollutants and toxins
 - C. drugs and body fluids
 - D. pollutants and body fluids
- C** 2. The two general types of illegal psychoactive drugs are _____ that can cause drowsiness and _____ that can cause hyperactivity.
- A. poisons and toxins
 - B. narcotics and opiates
 - C. depressants and stimulants
 - D. barbiturates and neurotransmitters
- C** 3. Before a drug can have its desired effect, it must be absorbed by the
- A. liver
 - B. nose
 - C. blood
 - D. kidneys
- A** 4. Drugs are broken down either in the region they affect or in the
- A. liver
 - B. brain
 - C. blood
 - D. kidneys

- D** 5. Most psychoactive drugs target cells within the
- A. nose and lungs
 - B. blood and urine
 - C. liver and kidneys
 - D. brain and spinal cord
- D** 6. What do psychoactive drugs change or mimic in the human body?
- A. Blood
 - B. Neuron
 - C. Synapse
 - D. Neurotransmitters
- C** 7. An individual who has consumed a depressant typically feels
- A. alert, dizzy, and hyperactive
 - B. clumsy, euphoric, and anxious
 - C. drowsy, relaxed, and pain-free
 - D. sleepy, hungry, and sexually aroused
- D** 8. Which of the following pairs of drugs are stimulants?
- A. Hashish and heroin
 - B. Alcohol and nicotine
 - C. Ecstasy and marijuana
 - D. Cocaine and amphetamines
- B** 9. Illegal drug use has been linked to high levels of
- A. fraud and abductions
 - B. property crime and petty theft
 - C. sexual assaults and armed robberies
 - D. motor vehicle thefts and traffic accidents
- A** 10. In the past thirty years, levels of drug-related arrests have
- A. tripled
 - B. doubled
 - C. been cut in half
 - D. remained the same

Value
2

Part B: Numeric-Response Questions

Body Parts and Fluids in which Psychoactive Drugs are Found:

1. CNS
2. Liver
3. Urine
4. Blood
5. Kidneys

①

1. The correct sequence of body parts and fluids, as numbered above, through which psychoactive drugs travel when taken by injection is 4 , 1 , 2 , 5 , and 3 .

Physiological Effects of Psychoactive Drug Abuse:

1. Confusion, insomnia, and impotence
2. Inflammation of heart lining and skin abscesses
3. Disruption of blood flow and increased chance of infection
4. Impairment of one's ability to operate a motor vehicle

①

2. Match each physiological effect numbered above with the psychoactive drugs given below.

Physiological effects: 1 3 4 2
Psychoactive drug: cocaine opiates marijuana amphetamines

Value
11

Part C: Written-Response Questions

Provide appropriate answers to the following. Explanations are best given in sentences.

③

1. a. Identify three body parts and/or fluids in which forensic toxicologists are most likely to find psychoactive drugs or their breakdown products?

Response should include any three of the following:

sweat, blood, urine, CNS, liver, or kidneys.

②

- b. Psychoactive drugs can also be detected in saliva. Explain why saliva samples are not used regularly by law enforcement officials to detect drug impairment in drivers.

Response should include any two of the following:

- ***No reliable roadside measurement device exists.***
- ***Under present laws, saliva analysis is inadmissible as evidence in court.***
- ***Test results are not as reliable as those for blood and/or urine. Too many false positive identifications result.***
- ***Present saliva tests cannot conclude the quantity of drug—only its presence.***

2. Compare and contrast illegal depressants and stimulants in the following ways:

②

- a. One way that each affects neural activity

Either of the following answers is acceptable for 1 mark.

- ***Some depressants prevent the release of neurotransmitters, which stops neural stimulation.***
- ***Some depressants mimic neurotransmitters that prevent pain.***

Either of the following answers is acceptable for another mark to bring the total up to 2.

- ***Some stimulants prevent the breakdown of neurotransmitters, causing neurons to fire continuously.***
- ***Some stimulants mimic the effects of certain neurotransmitters, causing neurons to fire continuously.***

②

- b. The general symptoms caused by each

- ***The general symptoms caused by depressants include drowsiness, sleep, insensitivity, and relaxation. (1 mark)***
- ***The general symptoms caused by stimulants include increased alertness, increased metabolism, hyperactivity, stimulation of sexual arousal, and prevention of hunger. (1 mark)***

②

3. State one possible reason drug-testing devices are not widely used by Canadian law enforcement agencies today.

Any of the following answers is acceptable for 2 marks:

Drug-testing devices are not widely used by Canadian law enforcement agencies today for several reasons:

- ***They are time consuming because drug-testing of urine and /or blood must be done in a hospital setting. They cannot be done on the roadside or at a police station.***
- ***No legislation currently exists that allows for roadside drug-testing.***
- ***The results are not 100% accurate. Some false positive results do occur.***
- ***Suspects cannot be forced to take a drug test. They must do so voluntarily.***

Total
29

Lesson 2 Assignment: The Effects of Poisons and Toxins

Value
10

Part A: Multiple-Choice Questions

Decide which choice best completes the statement or answers the question. Print the letter that corresponds to your choice on the blank next to the question.

- B** 1. Permanent damage or death often occurs when a poison targets the
- A. brain or muscles
 - B. brain or spinal cord
 - C. stomach or muscles
 - D. stomach or spinal cord
- D** 2. Toxins are
- A. deadly compounds produced by some plants
 - B. deadly toxic compounds produced by animals
 - C. synthetic compounds produced by drug companies
 - D. poisonous compounds produced in living organisms
- D** 3. Venom is
- A. drug from a plant
 - B. drug from an animal
 - C. toxin from a plant
 - D. toxin from an animal
- C** 4. Carbon monoxide causes suffocation by preventing oxygen from binding to
- A. ATP
 - B. enzymes
 - C. hemoglobin
 - D. mitochondria
- A** 5. If a person is given small doses of arsenic over time, his or her symptoms may be mistaken for
- A. food poisoning
 - B. accidental suicide
 - C. product tampering
 - D. intentional poisoning

- B** 6. Cyanide ingestion causes quick death because it targets cells in the
- A. lungs and liver
 - B. brain, spinal cord, and heart
 - C. hemoglobin and red blood cells
 - D. stomach, medulla oblongata, and muscles
- A** 7. Both arsenic and cyanide cause death by preventing the production of
- A. ATP
 - B. oxygen
 - C. hemoglobin
 - D. mitochondria
- D** 8. As a result of strychnine poisoning, death occurs soon after the poison begins to inhibit the function of the
- A. liver
 - B. blood cells
 - C. spinal cord
 - D. medulla oblongata
- C** 9. Most poisonings in North America are
- A. intentional suicides by adults
 - B. intentional homicides in teenagers
 - C. accidental poisonings in children
 - D. accidental suicides by senior citizens
- A** 10. Most deaths from poisoning in North America are
- A. intentional suicides by adults
 - B. intentional homicides in teens
 - C. accidental poisonings in children
 - D. accidental suicides by senior citizens

Value
2

Part B: Numeric-Response Questions

Use the following list to answer the next question.

Descriptions of Four Groups of Substances

1. harmful compound that is produced in a living organism
2. produced by animals and cause its effect through injection
3. chemical compound causing harmful effects after being absorbed into a body
4. products relief or prevents the effects of a harmful substance

①

1. Match the descriptions shown above with the names given below

Definition:	<u>3</u>	<u>1</u>	<u>2</u>	<u>4</u>
Substance:	poison	toxin	venom	antidote

Use the following list to answer the next question.

Deadly Effects Caused by Various Poisons

1. inhibits ATP production in the brain, spinal cord, and heart
2. prevents oxygen from binding to hemoglobin in red blood cells
3. after intense muscle spasms, stops activity in the medulla oblongata
4. causes multi-organ failure by preventing enzyme and mitochondrial activity

①

2. Match each description shown above with the poisons given below.

Effects:	<u>4</u>	<u>1</u>	<u>3</u>	<u>2</u>
Poison:	arsenic	cyanide	strychnine	carbon monoxide

Value
17

Part C: Written-Response Questions

Provide appropriate answers to the following. Explanations are best given in sentences.

②

1. a. Describe the harmful effects poison can have on the body.

Any two of the following:

- inhibit chemical reactions in the body
- organ damage
- inhibit activity in medulla oblongata
- blocks amino acid receptors in brain and spinal cord
- inhibit enzyme activity
- inhibit ATP production

- brain damage, birth defects, weight loss, sluggishness, vomiting, skin rash, abnormal heart beat
- prevents oxygen from attaching to the hemoglobin molecules within red blood cells.
- burning in throat

- nausea
- stomach cramps
- muscle spasms
- stop the heart and lungs
- increases hydrogen peroxide production

②

- b. Identify the two main ways poisons tend to be absorbed into the body.

1 mark each

- ingestion and/or eating
- inhalation and/or breathing

③

2. Explain why poisoning deaths do not always involve the consumption of true poisons.

Deaths due to poisonings do not always involve the consumption of true poisons. Rather they might be caused by the accidental or intentional overdose of a substance not actually considered a poison (such as Aspirin, alcohol, or household cleaners). (2 marks)

③

3. Discuss why homicides caused by intentional poisonings are rare.

- the most potent and easy-to-disguise poisons are not readily available to average consumer. 1 mark
- Intentional poisonings take much planning and preparation because the killer requires some knowledge of the precise amount of poison necessary to cause death. (1 mark).

②

4. a. Why is the actual number of poisonings in North America unknown?

The actual number of poisonings is unknown because not all cases of poisonings are detected or reported.

③

- b. How does the cause of reported poisonings compare to the cause of poisonings causing death?

The majority of reported poisonings are accidental overdoses in children (1 mark)

The majority of poisonings causing death are intentional suicides (1 mark)

①

- c. What substance is the most frequent cause of reported poisonings?

Most frequent cause of reported poisonings = household cleaners (1 mark)

①

- d. What substance is the most frequent cause of poisonings resulting in death?

Most frequent cause of death by poisoning = antidepressant drugs (1 mark)

Total
22

Lesson 3 Assignment: Forensic Toxicology Testing Techniques

Value
8

Part A: Multiple-Choice Questions

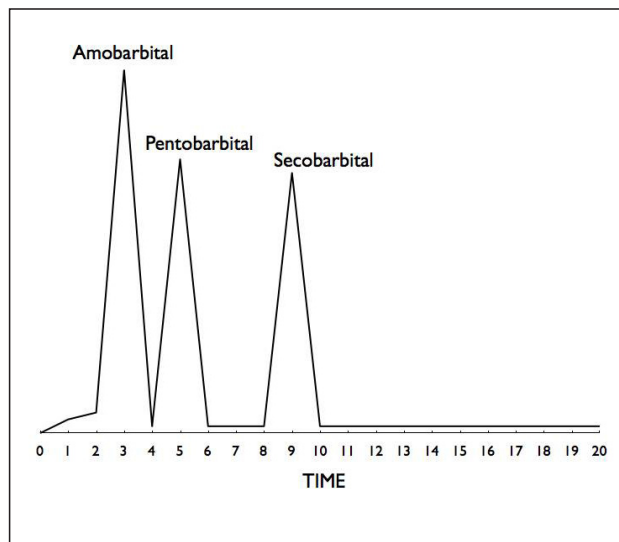
Decide which choice best completes the statement or answers the question. Print the letter that corresponds to your choice on the blank next to the question.

- D 1. The three most common cases that forensic toxicologists investigate are
- A. assaults, armed robberies, and fraud
 - B. attempted homicides, homicides, and suicides
 - C. poisonings, property crimes, and sexual assaults
 - D. alcoholic poisonings, drug-impaired driving, and drug overdoses
- A 2. Two examples of tissues or fluids that are never analyzed to determine the presence of drugs are
- A. tears and skin
 - B. saliva and sweat
 - C. blood and urine
 - D. bone and liver
- A 3. Which of the following pairs of toxicological screening techniques do not indicate the quantity of the drug or poison present?
- A. Colour testing and microcrystalline testing
 - B. Gas chromatography and mass spectrometry
 - C. Acid-base extraction and immunoassay testing
 - D. Radioimmunoassay and enzyme immunoassay
- C 4. Immunoassay testing involves reactions between
- A. blood and urine
 - B. enzymes and isotopes
 - C. antibodies and antigens
 - D. electrons and fragmentation patterns

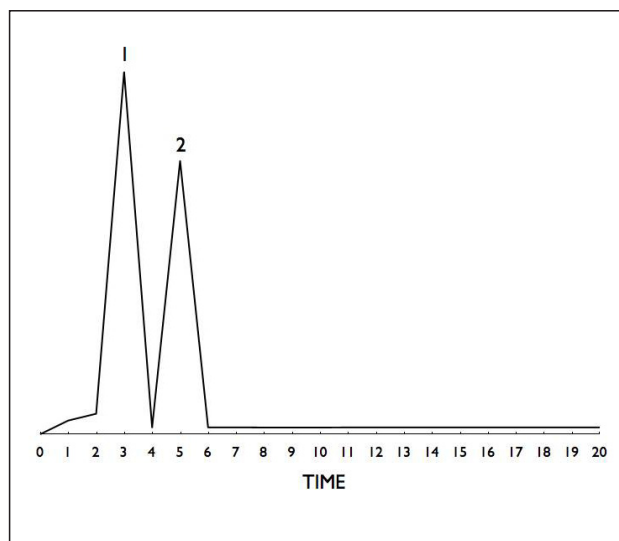
- A** 5. The height of a peak in a chromatogram can be used to determine the _____ of a certain _____ in an isolated sample.
- A. quantity, drug
 - B. quantity, antigen
 - C. identity, drug
 - D. identity, antigen
- A** 6. The most common confirmation technique used by forensic toxicologists is a combination of
- A. gas chromatography and spectrometry
 - B. colour testing and microcrystalline testing
 - C. acid-base extraction and immunoassay testing
 - D. radioimmunoassay and enzyme immunoassay
- B** 7. What tissues or body parts of Alexander Litvinenko were severely damaged as a result of being poisoned with polonium-210?
- A. Blood and urine
 - B. Liver and spleen
 - C. Brain and kidneys
 - D. Bone and stomach

Use the following pair of chromatograms to answer the next question.

Example Chromatogram of Various Barbiturates



Chromatogram from a Criminal Suspect



- B** 8. According to the chromatograms above, what barbiturates has the suspect taken?
- A. Amobarital and secobarbital
 - B. Amobarital and pentobarbital
 - C. Pentobarbital and secobarbital
 - D. Amobarbital, pentobarbital, and secobarbital

Value
2

Part B: Numeric-Response Questions

Use the following list to answer the next question.

Various Steps in Gas Chromatography

1. A chromatogram is created using a computer.
2. Isolated sample and gas are injected into tubing.
3. Diffusion of isolated sample and gas into a liquid.
4. Isolated sample is placed in heated injection chamber.
5. Chemical components of isolated sample enter a detector.

①

1. The correct sequence of steps involved in gas chromatography, as shown above, are 4, 2, 3, 5, and 1.

Use the following list to answer the next question.

Some Descriptions of Toxicology Testing Techniques

1. A chemical reaction may occur to produce a crystalline precipitate.
2. Small test strips are placed in a urine sample and a change in colour occurs.
3. A fragmentation pattern is created from substance's unique chemical composition.
4. Drug-specific antibodies are added to an isolated sample and a reaction is observed.

①

2. Match each description shown above with the correct toxicological testing technique given below.

Descriptions: 4 1 2 3
Testing Techniques: immunoassay microcrystalline colour testing mass spectrometry

Value
12

Part C: Written-Response Questions

Provide appropriate answers to the following. Explanations are best given in sentences.

③

1. Explain how the testing for drugs or poisons from a living person differ from the testing for drugs or poisons from a dead person.

Blood, hair, saliva, sweat, and urine may be analyzed in a living suspect or victim. (1 mark)

All of the above as well as bone, brain tissues, liver tissues, and stomach contents can be analyzed in a dead suspect or victim. (1 mark)

⑥

2. Explain three reasons the initial screening for drugs or poisons must be followed by confirmation testing. Indicate which tests have these problems.

• *Colour testing and microcrystalline testing do not indicate the quantity of the suspected substance. (2 marks)*

• *Colour testing and microcrystalline tests do not exist for all the drugs, poisons, and toxins. (2 marks)*

• *In immunoassay testing and gas chromatography testing, some organic substances have similar chemical structures to certain drugs, toxins, and poisons. These substances may produce false positive results. (2 marks)*

③

3. Why is mass spectrometry considered the most accurate method of confirmation testing in forensic toxicology?

• *Every drug, toxin, or poison produces a distinct fragmentation pattern according to its unique individual chemical structure. (1 mark)*

• *No two patterns are exactly alike. (1 mark)*

Total
26

Lesson 4 Assignment: Crime Case Studies Involving Forensic Toxicology

Value
12

Part A: Multiple-Choice Questions

Decide which choice best completes the statement or answers the question. Print the letter that corresponds to your choice on the blank next to the question.

Case Study #1: The Poisoning of a Politician

- D 1. Doctors think dioxin was placed in Viktor Yushchenko's food because he
- A. was barely unable to walk
 - B. was given heavy painkillers
 - C. had painful blisters on his skin
 - D. suffered from severe abdominal pains
- C 2. Dioxins are highly toxic chemical compounds produced in small concentrations when organic substances are burned in the presence of
- A. plastics
 - B. tobacco
 - C. chlorine
 - D. pesticides
- B 3. What symptoms of dioxins poisoning was visible even after Viktor Yushchenko had been given hospital treatment?
- A. Diabetes
 - B. Chloracne
 - C. Liver damage
 - D. Abdominal pain
- A 4. Some of the short-term effects of dioxin poisoning include
- A. chloracne, vomiting, and abdominal pain
 - B. diabetes, liver damage, and heart disease
 - C. reproductive organ damage, nausea, and cancer
 - D. inability to walk, painful blisters, and stomach ulcers

Case Study #2: The Jonestown Tragedy

- D** 5. During his 'white nights', Jim Jones had all his followers drink Kool-aid that they were tricked into thinking contained
- A. drugs
 - B. blood
 - C. venom
 - D. poison
- D** 6. Why type of poison was used to kill Jim Jones' followers?
- A. Valium®
 - B. arsenic
 - C. strychnine
 - D. potassium cyanide
- B** 7. In addition to poison, what did the Kool-aid contain that Jim Jones gave his followers?
- A. dioxin
 - B. Valium®
 - C. cocaine
 - D. arsenic
- A** 8. The poison used to kill the Jonestown followers causes death because it blocks the body's production of
- A. ATP
 - B. water
 - C. oxygen
 - D. mitochondria

Case Study #3: A Drug Overdose or an Intentional Poisoning - That is the Question

- D** 9. Evidence at the crime scene that suggests the victim died from a drug overdose was
- A. amobarbital and cocaine found in blood stream
 - B. strychnine and secobarbital found in blood stream
 - C. barbiturates, rat poison residue, and empty pesticide bottles
 - D. marijuana cigarettes, crack residue, and empty prescription bottles
- C** 10. What type of physical evidence was found near the victim's body?
- A. Rat poison
 - B. Fingerprints
 - C. Suicide note
 - D. Used syringe
- B** 11. What two pieces of physical evidence suggested the victim could have been poisoned?
- A. Fingerprints and rigor mortis
 - B. Used syringe and empty rat poison
 - C. Puncture wound and marijuana cigarettes
 - D. Crack residue and empty prescription bottles
- B** 12. What type of forensic toxicology test generated the chromatogram of the victim's blood (sample evidence #2)?
- A. Acid-base extraction
 - B. Gas chromatography
 - C. Immunoassay testing
 - D. Microcrystalline testing

Value
3

Part B: Numeric-Response Questions

Case Study #1: The Poisoning of a Politician

Use the following information to answer the next question.

Various Dioxin Poisoning Symptoms

1. cancer
2. nausea
3. diabetes
4. vomiting
5. heart disease
6. liver damage
7. abdominal pain
8. swollen pancreas

(Record your four-digit answer in descending order.)

①

1. Some of the long-term effects that Viktor Yushchenko could suffer from because of the dioxin he ingested include 6 , 5 , 3 , and 1 .

Case Study #2: The Jonestown Tragedy

1. coma
2. heart attack
3. convulsions
4. loss of consciousness

(Record your four-digit answer in descending order.)

①

2. The correct sequence of events, as numbered above, that each Jonestown poisoning victim would have experienced before death is 4 , 3 , 1 , and 2 .

Case Study #3: Was it a Drug Overdose or an Intentional Poisoning - That is the Question?

Use the following information to answer the next question.

Various Types of Drugs and Poisons

1. arsenic
2. cocaine
3. marijuana
4. strychnine
5. amobarbital
6. secobarbital
7. pentobarbital

(Record your four-digit answer in descending order.)

①

1. The chromatogram of the victim's blood (sample evidence #2) indicates that he had ingested 7 , 4 , and 3 .

Value
11

Part C: Written-Response Questions

Provide appropriate answers to the following. Explanations are best given in sentences.

Case Study #1: The Poisoning of a Politician

②

1. a. Explain why small amounts of dioxin are found in every human's blood.

Dioxins are found in a wide-range of common substances (i.e. food packaging, tampons, etc.) All people receive a small daily dose of dioxins; however, such small dosages do not seem to pose any health hazards. (2 marks)

①

- b. State how much dioxin Victor Yushchenko had in his blood stream as compared to the average individual.

Viktor Yushchenko had 1000 times the normal amount of dioxin in his blood. (1 mark)

Case Study #2: The Jonestown Tragedy

- ① 2. a. Whose visit to Jim Jones' colony in Guyana in November 1978 precipitated the mass suicide/mass murder?
US Congressman Leo Ryan, his aides, family members of some residents and various media visited Jonestown. (1 mark).
- ② b. Explain why this person paid a visit to Jim Jones' colony?
US Congressman Leo Ryan went to Guyana because allegations of human right abuses had been made. A possibility was that people were being held against their will in Jonestown. 2 marks

Case Study #3: Was it Drug Overdose or Intentional Poisoning? - That is the Question.

- ⑤ 3. Did the male victim in this case die due to a drug overdose or due to an intentional poisoning? Support your choice with four pieces of evidence.
Intentional poisoning (1 mark) - any 4 of the following are acceptable - 4 marks
- *eyes wide open*
 - *rigor mortis of legs and arms*
 - *thrashing about the room/struggle*
 - *suicide note can't be authenticated*
 - *syringe present even though crack cocaine, marijuana and barbituates don't use a syringe*
 - *rat poison found in dumpster (no fingerprints)*
 - *small puncture wound*

End of Assignment Booklet 1



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